Carrier-weighted psychometric networks: a label-free edge estimator that detects minority-carried structure invisible to pooled networks

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Abstract

A pooled edge correlation is an average over respondents: the same value can arise from weak coupling in everyone, strong coupling in a coherent minority, or accidental co-endorsement by scattered, mutually unrelated subsets. Pooled network pipelines cannot tell these apart. We introduce a label-free, edge-level estimator of carrier coherence, $w_e^* = |\rho_e|(1 - G_e)\max(\bar{O}_e, 0)$, whose key component $\bar{O}_e$ measures whether the respondents who carry an edge also carry its node-sharing neighbours. In a preregistered simulation with pooled correlations matched by calibrated design, the estimator separated minority-carried community edges from incoherent distractors with AUROC 0.985 versus 0.469 for raw correlations (paired difference +0.516, 95% CI $[+0.506, +0.526]$), while a concentration-only ablation remained at chance — confirming that coherence, not concentration, is the discriminating signal. End-to-end community recovery showed no improvement at $n = 600$; the preregistered diagnosis attributes this to candidate-edge selection, not the carrier statistic, and we report it as such.

1 Introduction

A correlation between two questionnaire items is an average over respondents. The estimate $|\rho| = 0.45$ is compatible with at least three different data-generating stories: (a) a weak coupling present in essentially every respondent; (b) a strong coupling present in a coherent 30% minority and absent elsewhere; and (c) no coupling in anyone, but incidental co-endorsement by scattered subsets of respondents who have nothing else in common. Standard psychometric network pipelines — regularized partial correlations or a maximally filtered graph, followed by community detection (Epskamp and Fried, 2018; Massara et al., 2016; Pons and Latapy, 2006; Golino and Epskamp, 2017) — operate entirely on the pooled value and are therefore blind to the distinction. This is an edge-level instance of the familiar warning that group-level statistics need not describe any individual: the ecological fallacy in its psychometric, non-ergodic form (Molenaar, 2004; Fisher et al., 2018).

The blindness has two concrete costs. First, communities carried by a coherent minority are attenuated at the pooling stage and are absorbed into neighbouring communities or dropped by sparsification, because each of their edges looks individually weak. Second, edges of type (c) — pooled correlations without any common carrier — pass every pooled screen and enter the network as if they were structure. Both errors leave no trace in pooled fit statistics: the network that results is internally consistent and wrong.

This paper contributes a label-free, edge-level estimator of carrier coherence that *detects* minority-supported structure invisible to pooled psychometric networks, while remaining a drop-in weight inside the standard bootEGA pipeline (Christensen and Golino, 2021). The estimator requires no grouping variable, no mixture model, and no change to the downstream community-detection machinery: it re-expresses each retained edge weight and nothing else.

What the estimator is not. Four boundaries prevent predictable misreadings. (1) It is not a mixture or latent-class network model: no partition of respondents is estimated, and the carriers of different edges may overlap, nest, or cross-cut freely. (2) It is not a group-comparison or subpopulation-similarity method (Christensen et al., 2023): no group labels are supplied at any stage. (3) It is not a meta-analytic between-study technique: the heterogeneity it measures is between respondents within a single sample. (4) It is not a multiple-comparison correction: it is an effect estimator, and inference is handled downstream by whatever resampling scheme the pipeline already uses.

2 The carrier-weighted estimator

2.1 Setup and carrier profiles

Let $x_{rj} \in \{0, 1\}$ be the response of respondent r to item j, and let ρ_e be the tetrachoric correlation for the item pair $e = (j, k)$. Define the per-respondent contribution of respondent r to edge e as

$$S_{re} \;=\; (x_{rj} - \bar{x}_j)\,(x_{rk} - \bar{x}_k), \tag{1}$$

the r-th summand of the covariance. The vector $S_{\cdot e} = (S_{1e}, \ldots, S_{ne})$ is the *carrier profile* of the edge: it records who supplies the association. Pooling collapses this vector to its mean; everything the present method does is done before that collapse.

2.2 Two carrier components

Two properties of the carrier profile carry the diagnostic information.

Concentration. The Gini coefficient G_e of the magnitudes $|S_{re}|$ measures how unevenly the edge's support is distributed over respondents. An edge carried by everyone has low G_e; an edge carried by a small subset — coherent or not — has high G_e. The even-support factor $(1 - G_e)$ down-weights concentrated edges.

Coherence. Let $N(e)$ be the set of edges that share a node with e — the neighbours of e in the line graph of the network. $\bar{O}_e$ is the mean cosine between the carrier profile of e and those of its neighbours:

$$\bar{O}_e \;=\; \frac{1}{|N(e)|} \sum_{f \in N(e)} \frac{\langle S_{\cdot e},\, S_{\cdot f} \rangle}{\|S_{\cdot e}\|\, \|S_{\cdot f}\|}. \tag{2}$$

Only node-sharing pairs enter the average — for $e = (j, k)$ that is $\deg(j) + \deg(k) - 2$ cosines, not all $\binom{|E|}{2}$ pairs — so the statistic is local and cheap even in dense networks; the structural reason for this restriction is given in Section 2.3. $\bar{O}_e$ asks the only question that concentration cannot: *do the same respondents carry the neighbouring edges?*

2.3 Why neighbour alignment: the wedge is the minimal detection unit

A single edge, considered alone, contains no information that discriminates scenario (b) from scenario (c) of the Introduction. The pooled ρ_e has already collapsed the respondent dimension. Concentration does not help either: a real minority-carried edge and an accidental co-endorsement are *both* concentrated, so G_e assigns them similar values. This is the a-priori reason to expect a concentration-only weighting to fail — a prediction the ablation in Section 3.4 tests directly.

The discriminating evidence lives one level up. Latent factors load on *nodes* (items), not on edges. If a factor F loads on item j, it leaves its trace simultaneously on every edge through j — (i, j), (j, k) — and, crucially, that trace passes through the *same* respondents, namely those in whom F is active. The shared node is the conduit of the common cause. A distractor edge's carriers, by contrast, have no structural reason to overlap with the carriers of any adjacent edge, so its expected alignment is zero.

The minimal unit of factor detection is therefore not the edge but the node-sharing edge pair — the *wedge*. Just as a correlation is a two-node statistic, carrier coherence is a two-edge statistic, one order up: $\bar{O}$ is a statistic on the line graph of the network, and the estimator's essence is to lift inference from node pairs to edge pairs. A community, in this view, is a connected set of high-coherence wedges sharing one latent cause, and carrier alignment is the empirical fingerprint of that sharing.

2.4 The full weight

The locked estimator combines the pooled magnitude with both carrier components:

$$w_e^* \;=\; |\rho_e|\,(1 - G_e)\,\max(\bar{O}_e,\,0). \tag{3}$$

The positive-part rule reflects the downstream contract: walktrap (Pons and Latapy, 2006) interprets a positive weight as attraction, so an edge whose carriers systematically *oppose* those of its neighbours ($\bar{O}_e < 0$) is removed rather than passed through as a small positive weight.

2.5 The minimal toy example: one triangle, two stories

If the wedge is the minimal detection unit, the natural worked example is the smallest structure made entirely of wedges: the closed triangle on three items i, j, k, whose three edges $e_1 = (i, j)$, $e_2 = (j, k)$, $e_3 = (i, k)$ form three node-sharing pairs, so that every edge's $\bar{O}$ is the mean of its two wedge cosines. Closing the triangle also closes a loophole a two-edge example leaves open: with an open wedge, the unused third pair (i, k) can still differ between scenarios, so a sceptic could ask whether triadic closure alone would have separated them. Here it cannot, by construction. Figure 1 shows two deterministic scenarios on $n = 18$ respondents (no random seed) in which *every* pooled 2×2 contingency table — all three edges, both scenarios — has the same four cells. All pooled pairwise quantities are therefore identical as a matter of arithmetic identity, not approximate matching: every ϕ coefficient is .25, every tetrachoric correlation .40, every Gini coefficient .29, and all item marginals coincide. Every S_{re} entry is a multiple of 1/9; the whole example can be verified by hand in a few minutes.

In scenario A one minority (respondents 1–3) endorses all three items and carries all three edges, while nine singleton respondents (three per item) endorse one item each, filling the off-diagonal cells. All three carrier cosines equal $+2/3$, so $\bar{O} = +2/3$ and every edge survives

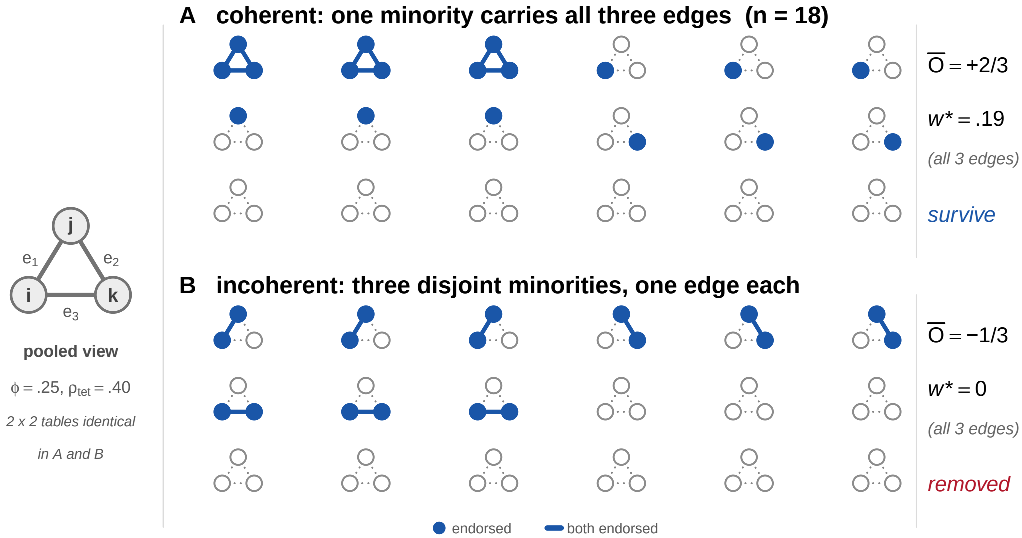

Figure 1: The minimal toy example: one closed triangle ($e_1 = (i, j)$, $e_2 = (j, k)$, $e_3 = (i, k)$), $n = 18$, two deterministic scenarios (no random seed) whose pooled 2×2 tables are *identical* on every edge — hence identical ϕ coefficients (.25), tetrachorics ($\rho_{\text{tet}} = .40$), Ginis ($G = .29$), and marginals, by arithmetic identity rather than approximate matching. Each respondent is drawn as their own copy of the triangle, its outline dotted: filled nodes mark endorsed items; an edge is drawn solid if and only if the respondent endorses both of its endpoints (the respondent *carries* that edge). The pooled view (left, drawn once) is identical for the two scenarios. **A**: one minority carries all three edges (nine singletons fill the off-diagonal cells); all carrier cosines are $+2/3$, $\bar{O} = +2/3$, every edge survives with $w^* = |\rho|\,(1 - G)\max(\bar{O}, 0) = .40 \times .71 \times 2/3 = .19$. **B**: three disjoint minorities carry one edge each; all carrier cosines are $-1/3$, $\bar{O} = -1/3$, and the gate $\max(\bar{O}, 0) = 0$ removes every edge — the first two factors of w^* are identical to A's. No pooled pairwise statistic — including triadic closure — distinguishes the panels; the carrier profiles do.

with $w^* = .19$. In scenario B the same pooled tables are produced by three *disjoint* minorities — respondents 1–3 co-endorse (i, j), 4–6 (j, k), 7–9 (i, k): every pairwise carrier cosine equals $-1/3$, so $\bar{O} = -1/3$ and all three edges are removed. No pooled pairwise quantity — correlation, concentration, marginal, or any function of them, including triadic closure — distinguishes the panels; the eighteen respondents do. (The full three-way $2 \times 2 \times 2$ table does differ, as it must: A has three respondents in the $(1, 1, 1)$ cell, B has none. The estimator recovers exactly this respondent-level information from pairwise objects, without ever forming the higher-order table.) Note that scenario B's edges are perfectly real *within their own carriers* — what the carrier profiles withhold is evidence of a *shared* cause, and that is exactly what the estimator prices. This point returns at scale as the oracle diagnostic of Section 3.4.

2.6 Known failure modes

Two limitations are stated before any results. If a community's loadings are *balanced* across two opposite-signed carrier groups, the pooled correlation itself vanishes and both pooled and carrier-weighted pipelines partially fail; a signed extension of $\bar{O}$ is future work, not part of the locked estimator. And the method addresses between-respondent heterogeneity in cross-sectional data;

within-person dynamics (Molenaar, 2004) are outside its scope. Both points return in Section 5.

3 Preregistered simulation

The simulation was preregistered before implementation, with the estimator, data-generating model, methods, outcomes, and decision rules locked in advance; one amendment, made before any confirmatory replicate was run, is described below and in Appendix A. The scientific question is the one the toy example poses: can the locked estimator distinguish a real community whose edges are carried by the same minority of respondents from equally strong pooled edges carried by mutually incoherent subsets?

3.1 Design

Fifteen binary items form three equal blocks. Blocks A and B are population-wide: for $j \in A$, $z_{rj} = \lambda F_{r,A} + \epsilon_{rj}$ with $\lambda = .75$ (standardized loading $\lambda^* = \lambda/\sqrt{\lambda^2+1} = .60$), and analogously for B. Block C is minority-carried: a single respondent indicator $M_r \sim \text{Bernoulli}(\pi_C)$, $\pi_C = .30$, governs every C item, $z_{rj} = \lambda_C M_r F_{r,C} + \epsilon_{rj}$ with $\lambda_C = 1.10$ (standardized loading .74 *within* carriers; the unstandardized $\lambda_C > \lambda$ compensates the π_C dilution, and the pooled within-C correlation nonetheless remains below the within-A/B level), so that all ten within-C edges share one carrier subgroup. Six distractor pairs, fixed in advance and spanning blocks, each receive a pair-specific factor active only in its own disjoint respondent subset — matched pooled correlation with *no* shared carriers. Latent responses are dichotomized at fixed, block-balanced thresholds. The distractor loading δ is calibrated in a discarded pilot (seed 20260718) so that the mean pooled $|\rho|$ of the distractors matches the within-C mean, then frozen. Figure 2 summarises the design.

One amendment was required at the pilot stage. The preregistered zero-mean pair factor $D \sim N(0,1)$ saturates under dichotomization: only half of a distractor's subset can ever reach the concordant-endorsement cell, capping the achievable pooled correlation below the calibration target for every δ. The amendment — made before any confirmatory replicate, under the preregistration's own revision rule — shifts the pair factor to $D \sim N(1,1)$ (an incoherent minority that jointly *endorses* both items, which is the scientific intent of the distractor) and raises π_D from .10 to .15, the prevalence at which the target becomes reachable. With these values, $\delta = 3.75$ achieved calibration to within .001 and was frozen. The confirmatory run used $n = 600$ and 1,000 replicates (seeds 202608250001–202608251000), with a 1,000-replicate no-structure control (202608260001–202608261000); no replicate failed.

3.2 Two evaluation levels

The production pipeline first selects candidate edges with TMFG on raw $|\rho|$ (Massara et al., 2016) and only then reweights retained edges; carrier weighting cannot rescue an edge TMFG has already removed. The preregistration therefore mandates two evaluations that may never be collapsed: an *estimator-level* test on a fixed candidate set containing all true within-block and distractor edges, and an *end-to-end* test (raw-$|\rho|$ TMFG $\rightarrow$ reweight $\rightarrow$ walktrap) with TMFG availability — the proportion of the ten true C edges retained — reported separately. One structural fact sharpens the second level: K_5 is non-planar, so a planar TMFG can retain at most nine of the ten within-C edges. Availability is capped at 0.9 by construction, and the question is how far below that cap the raw-correlation filter falls.

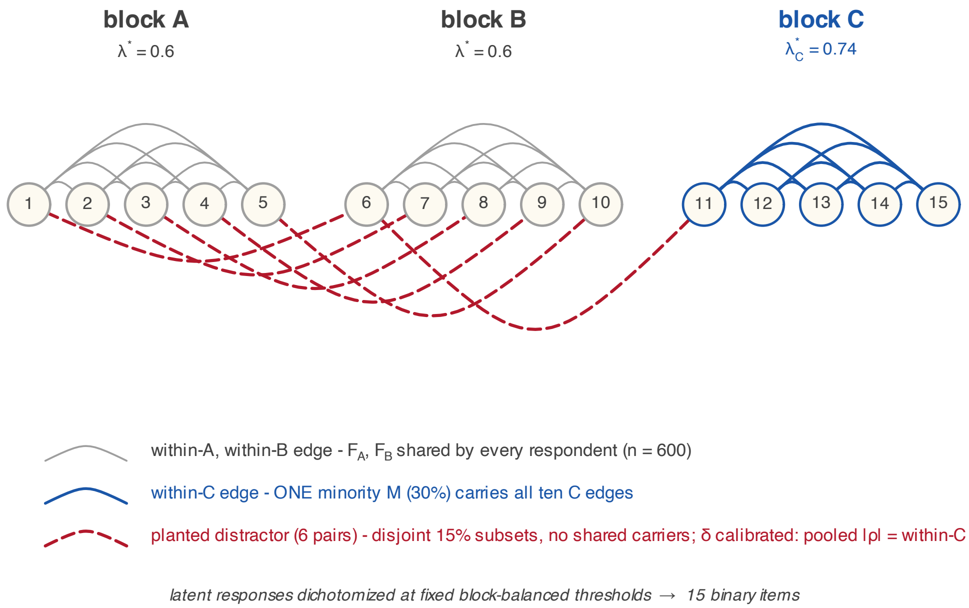

Figure 2: Generative design of the preregistered simulation. Fifteen binary items in three blocks of five; within-block edges are drawn as arcs above the item row, the six planted distractor pairs $(1,6), (2,7), (3,8), (4,9), (5,10), (6,11)$ as dashed arcs below. Blocks A and B are population-wide: every respondent ($n = 600$) shares F_A, F_B (standardized loading $\lambda^* = .60$). Block C is minority-carried: one subgroup M ($\pi_C = .30$) carries all ten within-C edges ($\lambda^*_C = .74$ within carriers). Each distractor pair receives a pair-specific factor active only in its own disjoint 15% respondent subset — no shared carriers — with loading δ calibrated in a discarded pilot so that mean pooled $|\rho|$ matches the within-C mean. The design scales the toy contrast of Figure 1: block C is scenario A (one coherent minority), the distractors are scenario B (incoherent minorities), and pooled correlation cannot separate them.

3.3 Methods and outcomes

Four deployable methods receive identical data: raw $|\rho_e|$; the Gini-only ablation $|\rho_e|(1 - G_e)$; the overlap-only ablation $|\rho_e| \max(\bar{O}_e, 0)$; and CW-full, Eq. (3). The ablations are mandatory because $(1 - G)$ rewards diffuse support and could in principle penalize the very minority concentration the method aims to recover. A fifth, non-deployable oracle computes correlations within the true carrier subgroup and serves as a diagnostic only. Estimator-level outcomes are the AUROC for ranking the ten C edges above the six distractors, the mean score separation, and the mechanism check (distributions of $|\rho|$, G, $\bar{O}$ for C versus distractor edges); end-to-end outcomes are ARI against the true partition, recovery of $k = 3$, exact recovery of block C, and TMFG availability. All comparisons are paired by replicate with percentile-bootstrap confidence intervals ($B = 10{,}000$).

The preregistration condenses these outcomes into four decision rules, locked before any confirmatory replicate together with the interpretation of every pass/fail pattern. Rule 1 is the estimator-level claim: CW must rank C edges above distractors better than raw correlation (AUROC CW $>$ raw). Rule 2 is the end-to-end claim: CW must improve *both* ARI and exact-C recovery over the raw pipeline. Rule 3 is the specificity control: in no-structure data, CW must not inflate false k=3 detections by more than .05 over raw. Rule 4 is the mechanism check: pooled $|\rho|$ must be matched between C edges and distractors (the calibration held), and the separation, if any, must be carried by $\bar{O}$ rather than by G. Results are reported below rule by rule.

3.4 Results

Rule 1 passed: AUROC 0.985 for CW versus 0.469 for raw, a paired-bootstrap difference of +0.516 with 95% CI [+0.506, +0.526]. Rule 2 *failed*: the ARI difference +0.0034 [+0.0001, +0.0067] excludes zero, but the exact-C difference +0.0020 [−0.0080, +0.0120] does not. Rule 3 passed: false k=3 rates were 0.295 (raw) and 0.308 (CW), a difference of +0.013. Rule 4 passed: pooled $|\rho|$ was matched (C .169, distractor .176), Gini did *not* separate the classes (.172 / .165), and $\bar{O}$ did (.102 / −.003).

Rule 4 is the wedge argument of Section 2.3 confirmed at scale (Figure 3): with pooled magnitude matched by design, concentration is indistinguishable between real minority edges and distractors — the Gini-only ablation achieves AUROC 0.464, chance — while coherence separates them cleanly, and overlap-only (0.985) matches CW-full (Figure 4). The oracle comparator returned an *inverted* AUROC of 0.002: within-carrier correlation is higher for distractor cliques than for C edges at the calibrated δ, because both mechanisms are perfectly real within their own carriers. Only cross-edge coherence distinguishes them — which is the thesis, so the oracle is reported as a diagnostic, not a competitor.

The end-to-end level tells a different and equally preregistered story. At n = 600 and π_C = .30 the raw pipeline already recovers the partition well, leaving little headroom: the per-replicate ARI distributions of the two pipelines nearly coincide (mean .946 raw versus .949 CW), and exact-C recovery is likewise indistinguishable (.815 versus .817; paired difference +.0020, CI spanning zero as reported above). The honest reading is a null. The failure map (Figure 5) locates the constraint precisely: exact-C recovery is near-ceiling (.99) whenever raw-$|\rho|$ TMFG retains its structural maximum of nine C edges, collapses to .37 below it, and does so *identically for every method* — overall exact-C (.817) tracks availability (.828) method-independently. Once the raw-correlation filter has dropped a true C edge, no reweighting of surviving edges can recover it. Under the preregistration's locked interpretation branch, this is "estimator succeeds, pipeline fails with TMFG availability as the bottleneck": the carrier statistic is viable, candidate-edge selection is the binding constraint, and carrier-aware edge selection is the subject of the next preregistered simulation. No post-result tuning was performed.

The preregistered secondary analyses — the 36-cell generalization grid over sample size, minority prevalence, and loading, and the bootstrap stability outcomes — are not reported here; the primary condition shows that end-to-end differences cannot emerge where raw recovery is already at ceiling, and the grid's informative region (smaller n, lower π_C) is deferred together with the carrier-aware selection study.

4 Empirical illustration

The estimator originated in a concrete measurement problem: the Gray–Wheelwright Jungian Type Survey, whose thinking–feeling axis is the least well established of its three (Mattoon and Davis, 1995, p. 228). In the pilot application that motivated this paper, replacing raw edge

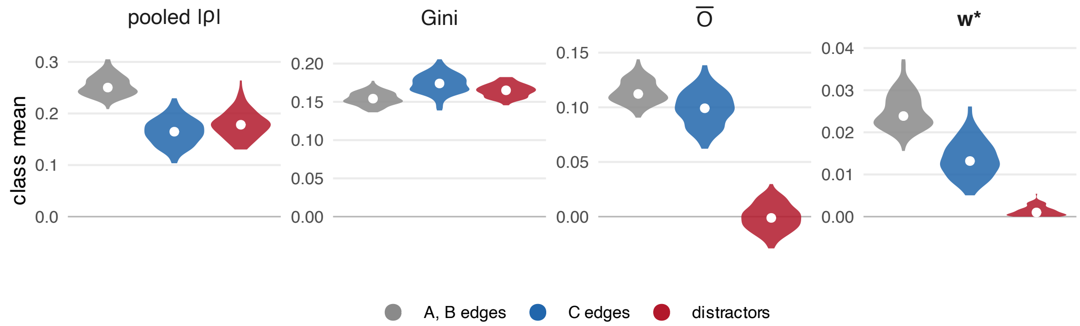

Figure 3: Mechanism check, all 36 structural edge classes (per-replicate class means over the 100 archived per-edge replicates; C and distractor values agree with the 1,000-replicate summary statistics). Pooled $|\rho|$ is matched between true minority-carried C edges and incoherent distractors by calibrated design, with both below the population-wide A/B level; Gini concentration separates none of the three classes; neighbour-alignment $\bar{O}$ alone isolates the distractors, while population-wide A/B edges and minority-carried C edges are both coherent. The resulting weight w^* (rightmost panel) preserves the A/B and C edges and drives the distractors to near zero: survival is decided by the coherence gate, not by magnitude or concentration. Class separability is quantified by the edge-level AUROC values reported in the text, not by significance tests on replicate means, whose p-values would reflect only the Monte Carlo budget.

weights with Eq. (3) inside an otherwise unchanged bootstrap EGA pipeline raised the bootstrap probability of recovering the theoretically expected three-community solution from 3.4% to 18.9% — precisely the behaviour expected if part of the axis structure is carried coherently by a minority of respondents and diluted by pooling. Substantive results on that instrument, including item selection and reliability, are reported in a separate manuscript (in submission) and are not claimed here; the illustration shows only that the estimator's target phenomenon occurs in real data.

5 Discussion

What the simulation establishes is deliberately narrow and, within that scope, unambiguous. At the estimator level, carrier coherence separates minority-carried structure from matched incoherent distractors essentially perfectly, the mechanism operates exactly as the wedge argument predicts, and the separation is attributable to the coherence factor alone: the Gini-only ablation sits at chance. What the simulation does *not* establish is an end-to-end recovery advantage at $n = 600$ — rule 2 failed, and the paper's title claims detection, not recovery, for that reason.

The ablation result also disciplines the formula itself. Overlap-only performed indistinguishably from CW-full, which licenses discussion — not post-hoc revision — of the $(1 - G)$ factor's role, and the honest reading is sharper than "neither helps nor harms." Analytically, G is a functional of the magnitude multiset $\{|S_{re}|\}$ alone and is therefore direction-blind: toy scenarios A and B have identical G by construction, so the factor cannot contribute to detection. Concentration of support is, moreover, the mathematical signature of *any* minority-carried edge, coherent or spurious, and the concentration information that matters in the network context is already priced — conditionally — by $\bar{O}$: misaligned concentration depresses the carrier cosines

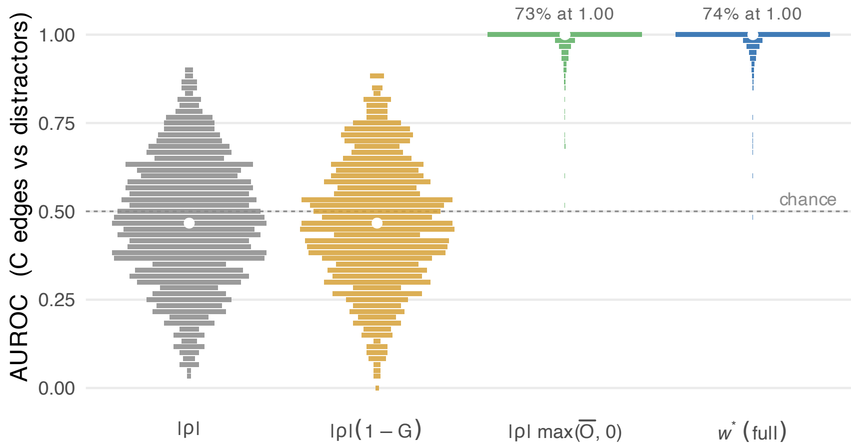

Figure 4: Ablation, estimator-level AUROC (C edges versus distractors) over 1,000 replicates; axis labels give each arm's scoring formula. AUROC over the 10×6 C–distractor pairs is discrete (grid 1/60), so each horizontal bar marks one attained value, with bar width proportional to the number of replicates at that value (normalized within arm); white dots mark medians. The Gini ablation $|\rho|(1-G)$ is at chance (mean .464); the coherence-gate ablation $|\rho|\max(\bar{O},0)$ carries the effect (.985); the full w^* matches it (.985) — the a-priori prediction of Section 2.3. The right-hand distributions collapse onto a single dominant bar at ceiling (73–74% of replicates at AUROC exactly 1.00, annotated) with a short tail below. The oracle subgroup comparator (mean AUROC .002, inverted) is reported in the text as a diagnostic.

through their normalization, while aligned concentration is precisely the signal. What $(1-G)$ adds beyond $\bar{O}$ is an *unconditional* penalty on minority-ness itself, in direct tension with the estimator's motivation. A deterministic dilution sequence, obtained by pencil from the toy construction and hence not tuned on confirmatory data, makes the tension concrete (Table 1): with the three-carrier scenario-A structure held fixed and only the all-zero background grown, both evidence terms strengthen monotonically while CW-full w^* is non-monotone and lowest exactly where the evidence is strongest, with $w^* \to 0$ as the coherent minority becomes rarer — the paper's own headline use-case. Nor does $(1-G)$ serve as an uncertainty proxy: it depends on the fraction m/n, whereas sampling precision scales with the absolute carrier count m, an inferential question that belongs to resampling downstream (§1), not to the point estimator. Equation (3) retains $(1-G)$ because the preregistration locked it; retention is preregistration discipline, not endorsement. Whether a Gini-free or a credibility-weighted (effective-carrier-mass) variant should replace it is the subject of the next preregistration, with the concentration information preserved in any case as a per-edge annotation (weight alongside carrier count and share). The preregistration's own interpretation table anticipated exactly this contingency.

Three limitations bound the claims. First, the balanced sign-flip scenario defeats the pooled correlation itself; a signed coherence statistic $(\bar{O}^+/\bar{O}^-)$ is future work. Second, the method addresses between-respondent heterogeneity in cross-sectional data and says nothing about within-person dynamics. Third — and most consequentially — the end-to-end analysis shows that the current pipeline's bottleneck is upstream of the estimator: raw-correlation TMFG discards true minority-carried edges that no subsequent reweighting can recover. Carrier-aware candidate-edge selection is therefore the next preregistered study, with the present results as its fixed

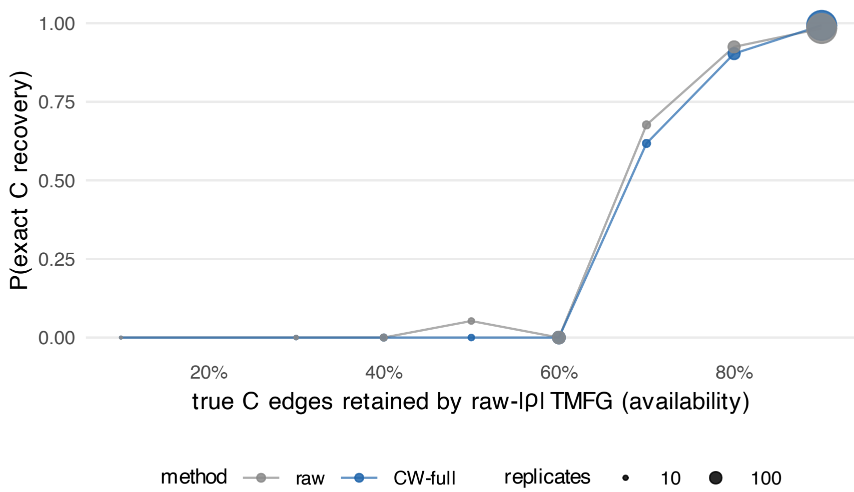

Figure 5: Failure map. Exact-C recovery as a function of the fraction of true C edges retained by the raw-$|\rho|$ TMFG (structural maximum 0.9; K_5 is non-planar). Recovery is near-ceiling at maximal availability and collapses below it identically for raw and CW: candidate-edge selection, not the carrier statistic, is the binding constraint.

Table 1: Deterministic dilution of the scenario-A triangle: the three carriers are held fixed while the all-zero background grows. Every value is an exact pencil computation from the toy construction (no simulation, no tuning). Both evidence terms strengthen monotonically; the $(1-G)$ factor drives the CW-full weight down exactly where the evidence is strongest, while the Gini-free product $|\rho|\max(\bar{O},0)$ is monotone in the evidence.

carriers	ρ_{tet}	$\bar{O}$	$1-G$	w^* (CW-full)	$\|\rho\|\max(\bar{O},0)$
3/18 (16.7%)	.397	+.667	.712	.188	.265
3/36 (8.3%)	.651	+.947	.379	.234	.617
3/60 (5.0%)	.737	+.985	.234	.170	.726
3/120 (2.5%)	.804	+.997	.121	.097	.802

benchmark. Companion work extending the carrier construction to antisymmetric (Hodge-type) flows and to permutation-based inference is cited here only as context and is developed elsewhere.

Reproducibility statement

The preregistration and its pre-confirmatory amendment are archived, date-stamped, in the project repository alongside the engine (`sim1_engine.R`), the analysis and figure scripts, and an md5 manifest of all outputs. All seeds are published (calibration 20260718, discarded; confirmatory 202608250001–202608251000; control 202608260001–202608261000), and the deterministic map from nominal seed IDs to RNG seeds is documented in the engine. The confirmatory run used R 4.5.1 with `psych` 2.6.5, `NetworkToolbox` 1.4.4, and `igraph`; figures were produced by a single script from the archived outputs. [TODO: public repository / sogum R-runner URL at submission].

References

Christensen, A. P., and Golino, H. (2021). Estimating the stability of psychological dimensions via bootstrap exploratory graph analysis: A Monte Carlo simulation and tutorial. *Psych*, 3(3), 479–500.

[TODO: verify exact reference: Christensen et al. (2023), subpopulation / group similarity in psychometric networks]

Epskamp, S., and Fried, E. I. (2018). A tutorial on regularized partial correlation networks. *Psychological Methods*, 23(4), 617–634.

Fisher, A. J., Medaglia, J. D., and Jeronimus, B. F. (2018). Lack of group-to-individual generalizability is a threat to human subjects research. *Proceedings of the National Academy of Sciences*, 115(27), E6106–E6115.

Golino, H. F., and Epskamp, S. (2017). Exploratory graph analysis: A new approach for estimating the number of dimensions in psychological research. *PLoS ONE*, 12(6), e0174035.

Massara, G. P., Di Matteo, T., and Aste, T. (2016). Network filtering for big data: Triangulated maximally filtered graph. *Journal of Complex Networks*, 5(2), 161–178.

Mattoon, M. A., and Davis, M. (1995). The Gray–Wheelwrights Jungian Type Survey: Development and history. *Journal of Analytical Psychology*, 40(2), 205–234.

Molenaar, P. C. M. (2004). A manifesto on psychology as idiographic science: Bringing the person back into scientific psychology, this time forever. *Measurement*, 2(4), 201–218.

Pons, P., and Latapy, M. (2006). Computing communities in large networks using random walks. *Journal of Graph Algorithms and Applications*, 10(2), 191–218.

A Preregistration deviations

One amendment, made before any confirmatory replicate was run and authorized by the preregistration's revision rule ("any revision creates Simulation 1b with a new file, seeds, and decision rules"):

Parameter	Preregistered	Amended (1b)	Reason
Pair factor $D_{r,d}$	$N(0,1)$	$N(1,1)$	zero-mean factor saturates under dichotomization (concordant-1 cell capped at $\pi_D/2$)
Distractor prevalence π_D	0.10	0.15	calibration target unreachable at 0.10 for any δ (δ-grid evidence archived in amendment file)
Confirmatory seed block	2026071800xx	2026082500xx	original block retired unused

The calibration target, decision rules, and all other parameters are unchanged. The original seed block was never executed; the amendment file records the full δ-grid from the discarded pilot.